

Distal Biceps and Triceps Tendon Ruptures

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DISTAL BICEPS TENDON RUPTURE

In 1941, Dobbie was the first to report the results of surgical reattachment of the distal biceps tendon. He recommended routine tenodesis of the biceps to the brachialis because of the risks associated with anatomic repair from the surrounding neurovascular structures within the antecubital fossa. Improved surgical techniques and a better understanding of the anatomy and biomechanics of the distal biceps tendon subsequently changed this sentiment. Many orthopedic surgeons now advocate early anatomic surgical repair of the distal biceps tendon in young, active patients.

Controversies continue to exist regarding the optimal treatment of distal biceps tendon ruptures. Many of these controversies focus on patient selection, approach to the radial tuberosity, and tendon fixation techniques. The first half of this chapter reviews the relevant pathoanatomy, workup, and treatment of patients with distal biceps tendon ruptures. The second half of the chapter focuses on triceps tendon rupture and repair.

Epidemiology

The estimated incidence of distal biceps tendon ruptures has risen from 1.2 to 2.55 per 100,000 population per year according to a recent epidemiologic analysis.^{1,2} Distal ruptures account for only 3% of biceps brachii tendon injuries.^{1,3} The injury is most commonly seen between the fourth and sixth decades of life, with an average age of approximately 50 years (range, 18 to 72 years).^{1,3-7} Several patient factors have been associated with distal biceps tendon ruptures, with the most significant factor being male sex. Eighty-six percent of distal biceps tendon ruptures occur in the dominant extremity, and they typically occur in highly active people, often with elevated body mass indices.^{1,2,8} Poisson regression analysis of tobacco use yields a 7.5 times increased risk of distal biceps tendon rupture among persons who smoke.^{1,4} Middle-aged men who use nicotine and anabolic steroids have a higher incidence of bilateral injuries.⁴

Anatomy

The biceps brachii muscle is contained in the anterior compartment of the arm and is composed of two heads. The long head originates from the supraglenoid tubercle within the shoulder joint, and the short head originates from the coracoid process. The two heads converge at the level of the deltoid tuberosity. The distal tendon gives rise to the lacertus fibrosus (bicipital

aponeurosis) before passing deep through the antecubital fossa and inserting onto the radial tuberosity. Anatomic investigation of the distal tendon reveals two distinct attachment sites, with a mean length of 92 mm and a mean width ranging from 2.9 to 6.1 mm.⁹ The tendon of the short head, which is more significant, attaches more distally on the tuberosity, and biomechanically acts as a flexor. The tendon of the long head inserts further from the axis of rotation of the forearm and acts as a strong supinator (Fig. 62.1).^{10,11} The lacertus fibrosus spreads out in an ulnar direction and blends into the forearm fascia, ultimately inserting onto the subcutaneous border of the ulna (Fig. 62.2). This aponeurosis may provide stability to the distal tendon.

Innervation of the biceps brachii is via the musculocutaneous nerve, a branch of the lateral cord of the brachial plexus. It penetrates the biceps muscle at an average of 134 mm distal to the acromion¹² and travels between the biceps and the brachialis before it penetrates the deep fascia of the arm and becomes the lateral antebrachial cutaneous nerve, which supplies sensation to the volar-lateral aspect of the forearm.

The biceps brachii is in close proximity to many vital neurovascular structures. The brachial artery, brachial vein, and the median nerve lie just medial to the biceps tendon and directly underneath the lacertus fibrosus. The brachial artery bifurcates into the radial and ulnar arteries at the level of the radial head. The radial recurrent artery arises from the radial artery and crosses laterally through the antecubital fossa, lying within the typical surgical field (Fig. 62.3). Lateral to the biceps tendon, the radial nerve enters the proximal forearm between the brachialis and the brachioradialis. The radial nerve divides into the deep and superficial branch anterior to the lateral humeral condyle.

Biomechanics

Owing to its dual insertion onto the radial tuberosity, the biceps muscle provides power and endurance for forearm supination and assists the brachialis in elbow flexion. The extent to which the biceps brachii contributes to elbow flexion correlates with the position of the forearm, with increasing contribution as the forearm is supinated. Furthermore, the biceps is able to exert maximum supination strength with the elbow at 90 degrees of flexion.

Morrey et al.¹³ performed a biomechanical study of 10 patients to evaluate the differences between conservative management

Abstract

Distal biceps and triceps tendon ruptures are functionally debilitating injuries that historically have been relatively rare among the general population. However, owing to increasing participation in competitive athletics, these injuries are becoming more frequent. In the absence of metabolic or systemic causes, these injuries typically occur in young, active individuals and are suffered when an eccentric load is placed on the tendon. Immediate pain, ecchymosis, and weakness are noted after the injury, with a palpable gap often becoming evident as the initial swelling subsides. The diagnosis is often clinical but can be greatly aided by magnetic resonance imaging. Once the diagnosis is established, tear chronicity, completeness, and location all play important roles in determining management. Although rarely selected, nonoperative intervention focuses on pain control, early range of motion, and strength optimization. Surgically, the ruptured tendon edges are repaired back to their native footprints using a variety of methods. Specifically for distal biceps, the surgical approach may vary as well, with proponents for both a one and two-incision technique. If repaired acutely, outcomes following surgical management of distal biceps and triceps tendon ruptures are excellent. Future directives should be aimed toward identifying those at risk for rupture and augmenting healing with the use of biologics.

Keywords

triceps
biceps
tendon
rupture
sports
elbow

and operative reattachment of distal biceps tendon ruptures. Immediate surgical fixation of the tendon to its insertion ultimately restored normal elbow flexion and forearm supination strength. In the group treated conservatively, an average loss of 40% supination strength and 30% flexion strength occurred.¹³ Baker and Bierwagen¹⁴ noted an 86% decrease in supination endurance in patients treated conservatively. Surgical management varies highly in the treatment of distal biceps injuries, both in terms of approach and fixation methods. Prud'homme-Foster et al.¹⁵ demonstrated that, when compared with a one-incision

approach, a two-incision approach allows for a more anatomic repair. In their cadaveric study, they demonstrated that forearms repaired anatomicly via a two-incision approach demonstrated 15% more supination torque at neutral pronosupination and 40% more supination torque at 45 degrees of supination compared with tendons fixed nonanatomicly via a one-incision approach.

Classification

Classification of distal biceps tendon ruptures is based on chronicity, degree of tear (partial vs. complete), and extent of retraction. The Ramsey classification uses these three characteristics to help guide treatment by predicting the ability to reattach the distal biceps tendon primarily to the radial tuberosity (Box 62.1).⁷

History

The mechanism of injury is nearly always a forceful, eccentric contraction of the biceps muscle. Examples of this mechanism include a preacher curl performed with excessive weight by a

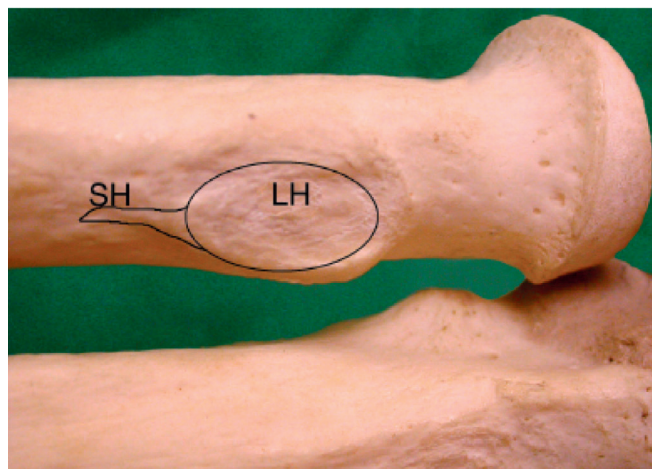


Fig. 62.1 Anatomy of the distal biceps tendon. Note the more distal insertion of the short head (SH), which acts as a strong flexor. The long head (LH), which inserts farther from the axis of rotation, acts as a strong supinator. (From Eames MH, Bain GI, Fogg QA, et al. Distal biceps tendon anatomy: a cadaveric study. *J Bone Joint Surg Am.* 2007;89:1044–1049.)

BOX 62.1 Classification of Distal Biceps Tendon Ruptures

Partial Ruptures

- Insertional
- Intrastance (elongation)

Complete Ruptures

- Acute (<4 weeks)
- Chronic (>4 weeks)
- Intact aponeurosis
- Ruptured aponeurosis

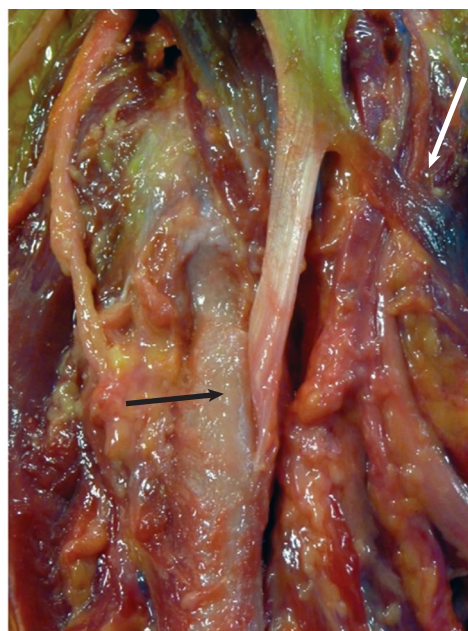
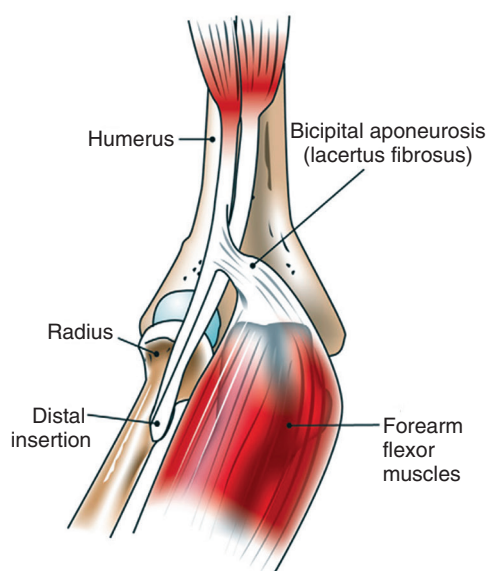


Fig. 62.2 Anatomy of the distal biceps tendon. The black arrow indicates the insertion of the distal biceps tendon; the white arrow indicates the bicipital aponeurosis. (From Miyamoto RG, Elser F, Millett PJ. Distal biceps tendon injuries. *J Bone Joint Surg Am.* 2010;92[11]:2128–2138.)

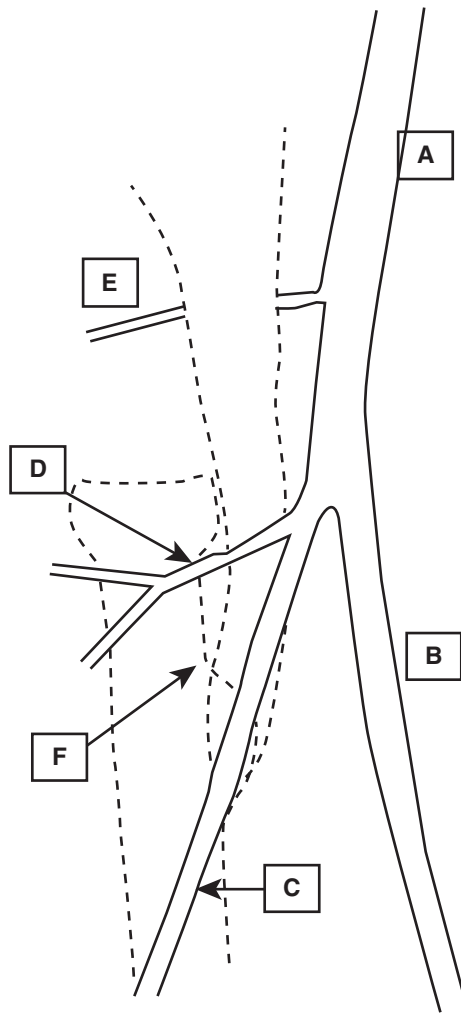


Fig. 62.3 Vascular anatomy surrounding the distal biceps insertion. Proximal radius and biceps tendon are depicted by dotted lines. A, Brachial artery; B, ulnar artery; C, radial artery; D, radial recurrent artery; E, dorsal radial recurrent artery; F, proximal most aspect of radial tuberosity. (From Zeltser DW, Strauch, RJ. Vascular anatomy relevant to distal biceps tendon repair. *J Shoulder Elbow Surg.* 2016;25[2]:238238.)

weight lifter and forcible extension of the elbow during an attempted tackle by a football player. The most common description and localization of initial pain is an abrupt, intense tearing sensation in the antecubital fossa. A palpable or audible “pop” is frequently recounted. Patients may report sudden and persistent weakness, especially with forearm supination and elbow flexion. As days pass and the swelling diminishes, the patient may notice a cosmetic deformity of the arm as the biceps retracts proximally. Many patients report the slow migration of skin discoloration from the hematoma within the elbow region toward the wrist over time. In patients with a partial biceps tear, the diagnosis is less clear. They may complain of a vague ache like discomfort within the antecubital region, which is exacerbated with forearm supination.

Physical Examination

Inspection of the elbow will often reveal deformity and proximal retraction of the biceps (termed a “Popeye muscle”), swelling,



Fig. 62.4 Inspection of an acute distal biceps tendon rupture with proximal muscle retraction and ecchymosis.

and ecchymosis within the antecubital fossa and medial aspect of the forearm (Fig. 62.4). Palpation elicits tenderness in the antecubital fossa and a defect of the tendon compared with the contralateral arm. In patients with an intact lacertus fibrosus, the deformity may be less pronounced. Weakness and pain with resisted forearm supination compared with the contralateral side is universally observed in complete ruptures.

Other physical examination tests specific for distal biceps tendon ruptures include the hook test and the biceps squeeze test. The hook test was originally described by O’Driscoll et al.¹⁶ and is performed by first placing the patient’s arm at 90 degrees of flexion. Next, the examiner attempts to hook the lateral edge of the biceps tendon with the index finger by pulling from a lateral to medial direction within the antecubital fossa. A positive test elicits an intact cordlike distal biceps tendon. The test is reported to have 100% sensitivity and between 85% and 92% specificity. One needs to be sure that they are feeling for the distal biceps tendon, which occasionally may be tricky as they palpate an intact lacertus fibrosus. The biceps squeeze test¹⁷ is similar to the Thompson test for Achilles tendon ruptures. The examiner squeezes the biceps brachii muscle, and in patients with an intact distal tendon, the forearm should supinate. Pain with palpation over the posterolateral aspect of the radial tuberosity while pronating the forearm is suggestive of distal biceps tendinopathy.

Imaging

A standard elbow radiographic trauma series (i.e., anteroposterior, lateral, and oblique views) should be obtained to evaluate for other causes of anterior elbow pain. Generally there are no signs of acute bony injury, but anterior soft tissue swelling may be appreciated. In rare cases an avulsion fracture of the radial tuberosity may occur and should be visible on radiographs.

Magnetic resonance imaging (MRI) is an excellent modality for assessment of the integrity of the distal biceps tendon, but it is not always necessary to make or confirm the diagnosis.¹⁸ MRI is valuable for differentiating complete from partial tears in equivocal cases, such as in obese patients and persons with an intact lacertus fibrosus. Positioning the arm for an MRI with the elbow flexed, shoulder abducted, and the forearm supination, the so-called “FABS view,” allows for complete imaging of the distal biceps tendon, typically on a single image (Fig. 62.5).¹⁹

Treatment Options

Nonoperative treatment may be considered especially for elderly, low demand individuals and those with partial tendon ruptures wishing to avoid surgery. The typical nonoperative protocol consists of a brief period of immobilization, analgesia, and physical therapy. The goals of physical therapy are the restoration of range of motion (ROM) with gradual elbow flexion and forearm supination strengthening. Patients who elect nonoperative treatment must be counseled on the risk of chronic activity-related pain, fatigue, loss of supination strength between 40% and 60%, and flexion strength of up to 30% and endurance.^{13,14,20,21} Patients also must be made aware that outcomes after delayed repair (>4 weeks) may be inferior to those obtained with acute repair and may require tendon grafting or be deemed irreparable.

Once the decision is made to treat a distal biceps tendon rupture surgically, several factors must be carefully taken into consideration prior to proceeding. These include patient factors such as occupation and function, history of anabolic steroids use, how much time has elapsed from the injury, how to expose of the radial tuberosity (one vs. two incisions) and how best fixate the tendon to bone. To date no single surgical plan has demonstrated clear superiority; operative treatment must be tailored to the individual patient and the proficiency of the surgeon.

Injury chronicity is perhaps the most important factor for the surgeon to consider. Early diagnosis may prevent tendon retraction permitting easier reattachment of the biceps tendon to the radial tuberosity regardless of the integrity of the lacertus

fibrosus. In this regard, primary reattachment is generally reliable if performed within 4 weeks of the injury. An intact lacertus fibrosus may minimize tendon retraction and increase the window for reattachment beyond 4 weeks. Concurrent rupture of the lacertus fibrosus will lead to proximal retraction of the tendon and formation of scar tissue with attachment to the adjacent brachialis muscle. Delayed surgery often necessitates extensive scar dissection to liberate the coiled tendon and regain critical length for primary reattachment.

Primary repair may not be possible in persons with chronic injuries, and tendon autograft/allograft reconstruction may be offered to these patients as an alternative to nonoperative care. The most commonly used autografts are semitendinosus and fascia lata, whereas the standard allograft is the Achilles tendon. Delayed reconstruction with use of a tendon graft historically has a less predictable outcome than early anatomic repair of the native tendon. Improvements in strength and endurance are possible, but delayed reconstruction may be reserved for symptomatic patients with significant disability.

Nonanatomic repair or distal biceps tenodesis to the brachialis muscle was first proposed in an effort to decrease complications associated with anatomic reattachment. Several studies have evaluated the efficacy of nonanatomic repair and have found excellent return of flexion strength but, predictably, little or no improvement in forearm supination strength.^{22,23} With modern anatomic reattachment techniques, the complication rate has decreased and nonanatomic repair has largely fallen from favor. One exception is a patient with a chronic, irreparable tear who has a chief complaint of activity-related pain and cramping rather than weakness and loss of endurance.

Much debate exists as to whether to perform a one or two incision repair technique. The discussion focuses on the ability to anatomically reattach the biceps tendon to the footprint of the radial tuberosity and the complications associated with each approach. Originally the surgery was performed using a one-incision technique to expose the radial tuberosity through the antecubital fossa. Given the relatively high rate of nerve injury,

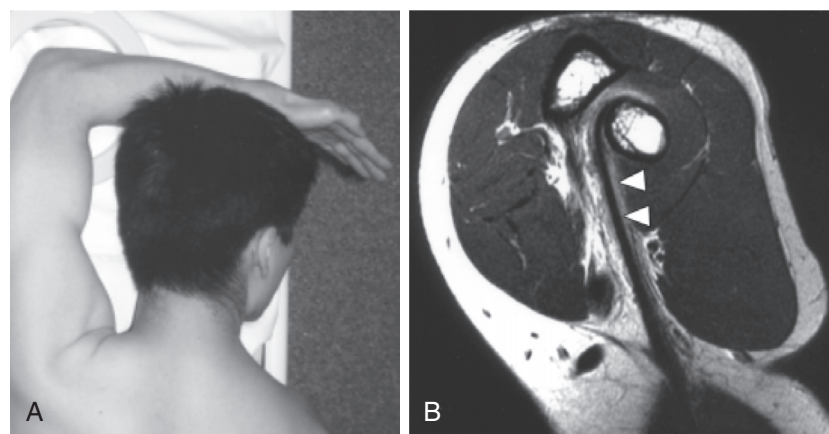


Fig. 62.5 Utilization of the flexed, abducted supinated “FABS view” during magnetic resonance imaging (MRI) for the detection of distal biceps injuries. (A) Patient positioning for the “FABS view”: the patient is prone on the MRI table, with the elbow flexed, the shoulder abducted and the hand supinated. (B) Proton density-weighted MR image (TR/TE 3,000/34) in the “FABS” position demonstrating the entire tendon on a single cut. White arrowheads, Abnormal signal within the tendon, indicating a partial tear.

Boyd and Anderson²⁴ developed a two-incision technique to expose the footprint through an additional posterolateral approach. Attachment of the biceps tendon through the posterolateral approach permits a more anatomic reconstruction to the radial tuberosity and requires less extensive anterior exposure, theoretically decreasing the risk of iatrogenic nerve injury. Furthermore, Hasan et al. demonstrated that through a single incision approach, only 9.7% of the repair site was within the original footprint, compared with 73.4% for the double incision approach.²⁵ Critics of this two-incision approach point to the unique risk of radioulnar synostosis. Failla et al.²⁶ responded to this drawback by reporting a case series of four patients with radioulnar synostosis who were treated with a modification of the Boyd and Anderson procedure involving a muscle-splitting approach between the common extensor mass and the supinator. This modification allows exposure of the radial tuberosity without violating the ulnar periosteum and decreases the risk of radioulnar synostosis.²⁶ A subsequent study revealed that the rate of radioulnar synostosis increases if the posterolateral incision is made over the subcutaneous border of the ulna rather than over the common extensors.²⁷

Four fixation techniques are presently being used to reattach the distal biceps tendon to the radial tuberosity: transosseous tunnels, suture anchors, interference screws, and cortical fixation buttons (or a combination thereof). Several biomechanical studies have compared fixation strength and stiffness between the different fixation techniques.^{28–33} In general, all fixation methods provide adequate fixation strength compared with the intact tendon, with the highest load to failure achieved using a cortical fixation button (Table 62.1).^{15,34}

Decision-Making Principles

Acute operative fixation of the distal biceps may be offered to patients who require strength and endurance with elbow flexion and forearm supination. Nonoperative treatment of complete ruptures will yield between 12% and 40% loss of supination strength, up to 86% loss of supination endurance, and consistently decreased Disabilities of the Arm, Shoulder and Hand (DASH) and European Society of Surgery of the Shoulder and Elbow scores. Conversely, early anatomic reconstruction of the distal biceps can restore elbow flexion and supination strength and endurance and demonstrates higher clinical and functional outcomes.^{13,14,20,21,35} When considering nonoperative treatment, the practitioner needs to consider the patient's functional status and needs to inform patients of the sequelae should they proceed with this option.

Postoperative Management

Rehabilitation protocols vary depending on the surgical technique and patient compliance. Early in the postoperative course, a period of immobilization or protective splinting is typical. Gentle passive ROM is permissible within the first few weeks. Some surgeons use a hinged brace to block terminal elbow extension initially, which is meant to protect the healing tendon from high tensile forces. After 3 weeks, patients may begin active assisted ROM, and full active motion is usually achieved 6 to 8 weeks after surgery. Resistive exercises are restricted until 8 weeks after surgery, at which time a gradual approach to biceps strengthening exercises is prudent to avoid a repeat tendon rupture. Although many patients prefer a home exercise regimen, it may be wise to refer noncompliant patients to a physical or occupational therapist for guidance and supervision during this phase. Most patients recover ROM and desired strength between 3 and 6 months after surgery.

Cheung et al.³⁷ evaluated early self-administered passive ROM exercises. After surgery, a hinged elbow brace locked at 90 degrees of flexion was immediately applied. On the first postoperative day, the brace was unlocked to allow passive ROM exercises from 60 degrees to full flexion with full forearm rotation. The elbow extension block was reduced at 2 weeks to 40 degrees, at 4 weeks to 20 degrees, and at 6 weeks to full extension. Strength training was begun at 8 weeks. With this protocol of immediate elbow motion following distal biceps tendon reattachment, no deleterious effects on healing or strength were observed.

Return to sport after distal biceps tendon rupture has not been specifically studied. The general consensus is that patients may return to unrestricted activities at 16 to 20 weeks after surgery. Athletes who participate in contact sports, such as football and rugby players, may need the full 20 weeks to permit tendon maturation and to prevent a repeat rupture.

Results

Several studies have evaluated the outcome of one- and two-incision distal biceps tendon repairs. Both techniques improve DASH scores and elbow flexion and forearm supination strength.^{27,38–43}

Karunakar et al.³⁸ reported on 21 patients after use of the modified Boyd and Anderson technique with bone tunnels. At 44 months all patients reported good to excellent results. Compared with the uninjured arm, endurance was decreased 38% in supination and 33% in flexion. A 15% rate of heterotopic ossification was noted.

TABLE 62.1 Methods for Distal Biceps Tendon Repair: Fixation Strength

Study	LOAD TO FAILURE (N)				
	Intact Tendon	Transosseous Tunnel	Suture Anchor	Interference Screw	Cortical Fixation Button
Berlet et al. ²⁸		307 (±142)	220 (±54)		
Lemos et al. ³¹		203	263		
Idler et al. ³²	204 (±76)	125 (±23)		178 (±54)	
Greenberg et al. ²⁹		177			584
Mazzocca et al. ³³		310	381	232	440



Authors' Preferred Technique

Distal Biceps Tendon Rupture

Our preferred technique is a single-incision anatomic repair of the distal biceps tendon using a combination of both a fixation button and an interference screw.³⁶ To perform this technique, a transverse incision is made anterior to the radial tuberosity with the arm in extension, usually 3 to 4 cm distal to the elbow flexion crease. An alternative approach is a longitudinal incision over the radial tuberosity, which is more extensible than a transverse approach. The antecubital veins are ligated or mobilized. The lateral antebrachial cutaneous nerve is identified lateral to the biceps course and carefully protected to avoid traction injury. To access the footprint of the radial tuberosity, the leash of radial recurrent artery is identified and ligated. At this point the ruptured tendon is either found locally (when the lacertus fibrosus is intact) or proximal blunt dissection is carried out if the tendon has retracted (i.e., when the lacertus fibrosus is torn). Along the original course of the tendon, large seromas are frequently discovered and evacuated. As the chronicity of the injury increases, a pseudosheath of fibrous scar tissue may mimic the actual tendon. Often this pseudosheath must be carefully débrided to unveil the native tendon end buried within this "cocoon." Once retrieved, many surgeons will place a clamp (e.g., an Allis clamp) at the terminal tendon and manually release proximal adhesions or even "water ski" with the tendon to achieve maximal excursion. To aid in this, the tourniquet may need to be deflated as this may tether the biceps muscle. Using a heavy nonabsorbable suture (e.g., No. 2 FiberWire, Ticron, or Ethibond), a whipstitch is woven through the distal biceps tendon from proximal to distal, leaving two long limbs exiting the terminus.

The radial tuberosity is exposed with the elbow in full extension and maximal passive supination to protect the posterior interosseous nerve. A retractor may be placed by the radial aspect of the tuberosity, but care must be exercised to avoid excessive lateral retraction to prevent injury to the posterior interosseous

nerve. A 3.2-mm guide pin is then drilled bicortically through the radial tuberosity. An acorn-shaped reamer is passed over the guide pin to create a unicortical tunnel. The suture limbs are passed through a fixation button, and the button is passed bicortically through the radial tuberosity. The tendon is advanced approximately 1 cm into the tunnel and the sutures are tied, securing the biceps tendon within the radial tuberosity (Fig. 62.6). An interference screw is placed into the tunnel, displacing the tendon more distally and ulnar to better recreate the anatomic insertion site.



Fig. 62.6 Anatomic repair of a distal biceps tendon. (From Sethi P, Cunningham J, Miller S, et al. Anatomical repair of the distal biceps tendon using the tension-slide technique. *Techniques Shoulder Elbow Surg.* 2008;9[4]:182–187.)

McKee et al.⁴² compared one-incision repairs with two suture anchors, using the uninjured arm as a control. At final follow-up, no difference was found in DASH scores. Strength was 96% in flexion and 93% in supination compared with the uninjured side. No differences in ROM were observed.

In a study evaluating outcomes after EndoButton (Smith and Nephew, Andover, MA) fixation, Peeters et al.⁴⁴ reported an average Mayo Elbow Performance Score of 94 points. Isokinetic testing showed 80% recovery of flexion strength and 91% recovery for supination. Of 26 patients, 2 experienced asymptomatic heterotopic ossification, and three had asymptomatic disengagement of the EndoButton.

Grewal et al.⁴⁵ performed a direct comparison between one-incision repairs using suture anchors and the modified Boyd and Anderson approach using bone tunnels. They found no difference in patient reported outcomes for pain and function at 2 years postoperatively. The one-incision cohort had greater improvement in flexion (142.8 vs. 131.1 degrees) but an overall higher complication rate, with the majority of complications being lateral antebrachial cutaneous nerve palsies. Similarly, Shields et al.⁴⁶ found a higher incidence of lateral antebrachial cutaneous neuropraxia following one-incision repairs, but found no difference in strength or ROM and equivalent patient reported outcomes. Schmidt et al.⁴⁷ investigated factors that determine strength after distal biceps reattachment and found⁴⁷ a two-incision approach to better restore strength than a one-incision

approach with no ultimate differences in forearm motion between groups.

Several fixation techniques may be used to affix the tendon to the radial tuberosity. Fixation techniques and respective load to failure are reviewed in Box 62.2. Techniques that increase the load to failure may allow earlier rehabilitation after surgery.

Chronic tears (i.e., those that occurred >6 weeks earlier) may require tendon grafting. Darlis et al.⁴⁸ retrospectively reviewed seven patients with chronic distal biceps tendon ruptures. These investigators sutured Achilles tendon allograft to the distal biceps tendon remnant and secured it to the radial tuberosity with two anchors. Mean elbow flexion was 145 degrees, and mean pronosupination was 170 degrees. Mean supination strength was 87% of the contralateral arm. Wiley et al.⁴⁹ compared nonoperative treatment versus operative fixation using a semitendinosus allograft in chronic distal biceps tendon ruptures. At final follow-up, flexion and supination strength returned to normal range in the operative group, and they returned to 20% of normal in the nonoperative group. Endurance in both groups was normal.

Complications

A systematic review of the literature evaluating one- and two-incision distal biceps tendon repairs revealed an 18% complication rate for one-incision repairs and a 16% complication rate for two-incision repairs. The majority of complications were nerve injuries (13%) after one-incision repair and loss of rotation and supination strength after two-incision repairs.^{38,41} Heterotopic

BOX 62.2 Factors Associated With Distal Triceps Tendon Ruptures

Activities

- Football
- Weight lifting

Steroid Use

- Anabolic steroids
- Locally injected corticosteroids

Metabolic Disease

- Diabetes mellitus (type 1)
- Chronic renal disease
- Renal osteodystrophy
- Hyperparathyroidism

Other

- Olecranon bursitis
- Marfan syndrome
- Osteogenesis imperfecta
- Systemic lupus erythematosus
- Rheumatoid arthritis

ossification has been reported to be more common following two-incision repairs.⁵⁰ More recently, a randomized controlled trial and a meta-analysis demonstrated significantly higher rates of neurapraxia of the lateral antebrachial cutaneous nerve following a single incision approach.^{50,51} Although a slight predilection to certain complications appears to exist for each technique, it is important to recognize that no technique completely eliminates the risk of any one complication.

DISTAL TRICEPS TENDON RUPTURE

Elbow flexion is generally regarded as the most essential function of the elbow because gravity can aid in extension. However, the lack of active elbow extension can compromise even the most basic activities of daily living, such as pushing up out of a chair from the seated position. The loss of elbow extension strength is most devastating for athletes of various sports. Therefore prompt recognition and treatment of distal triceps tendon ruptures is warranted.

Epidemiology

Distal triceps tendon ruptures are relatively uncommon injuries. However, their incidence is likely rising because the training of high-performance athletes and the use of performance-enhancing substances has increased over the past two decades.

Distal triceps tendon ruptures occur predominantly in a subset of high-risk people and are exceedingly uncommon in the general population. Strenuous activities, steroid use, and metabolic disease can predispose persons to this condition (see Box 62.2).^{52–56} Football players, particularly linemen and powerlifters, are at increased risk because of the nature of their sport and the intensity of their training.⁵⁷ Hyperparathyroidism, which is typically a result of chronic renal disease, is also associated with tendon ruptures. Hyperparathyroidism alters the enthesis

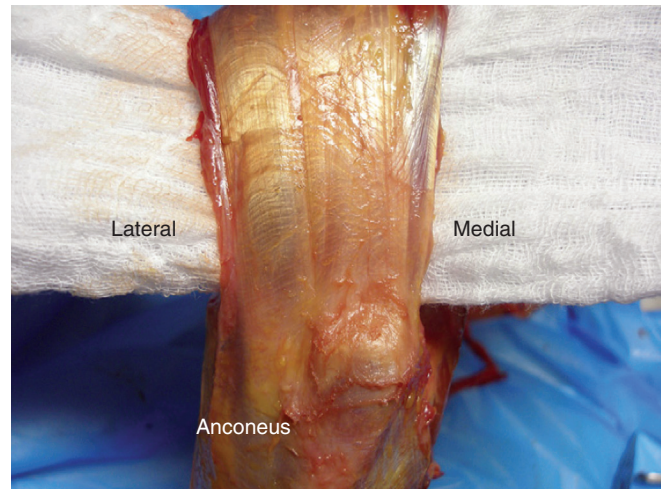


Fig. 62.7 Insertional anatomy of the distal triceps tendon. (From Keener JD, Chafik D, Kim HM, et al: Insertional anatomy of the triceps brachii tendon. *J Shoulder Elbow Surg.* 2010;19[3]:399–405.)

of all tendon insertions. The mechanism of insult is not entirely understood but is likely a result of osteoclast overstimulation, leading to the leaching of calcium from the tendon insertion. The enthesis is thus weakened and becomes prone to microfracture and avulsion injuries.

Anatomy and Biomechanics

The triceps brachii is a large muscle encompassing the entire posterior compartment of the arm. Innervation is entirely from the radial nerve (C6–C8). This muscle is composed of three heads: lateral, long, and medial. The medial and lateral heads originate from the posterior humerus and the intermuscular septum and act solely to extend the ulnohumeral joint. The long head originates from the infraglenoid tubercle and may also assist in shoulder adduction and extension.

Keener et al.⁵⁸ evaluated the insertional anatomy of the triceps and described a broad footprint on the olecranon measuring 466 mm.³ The triceps insertion can be divided into the triceps insertion proper and the lateral triceps expansion. Medially, the triceps attaches directly to the olecranon, forming the medial border of the triceps tendon proper. Laterally, the triceps tendon expands beyond the olecranon and becomes continuous with the anconeus and antebrachial fascia and inserts distally on the radial aspect of the proximal ulna (Fig. 62.7).

Classification

In classifying distal triceps tendon ruptures, three characteristics must be evaluated. First, the degree of tendon tear is an important factor in decision-making. Patients with complete tears and high-grade partial tears may be candidates for surgery, whereas low-grade partial tears often respond to conservative treatment. Second, the anatomic location of the tear will influence the choice of surgical technique and strength of fixation, with muscle belly and musculotendinous junction repairs being less stout than their tendinous counterparts. Lastly, the chronicity of the tear is an important factor in determining surgical feasibility and for preoperative planning (i.e., graft augmentation; see Box 62.3).

BOX 62.3 Classification of Distal Triceps Tendon Ruptures

Degree of Tear

- Partial
 - <50%
 - >50%
- Complete
 - Intact lateral expansion
 - Ruptured lateral expansion

Tear Location

- Muscle belly
- Musculotendinous insertion
- Insertional
- Olecranon avulsion fracture

Chronicity of Tear

- Acute (<3 weeks)
- Chronic (>3 weeks)

History

The diagnosis of distal triceps tendon ruptures may not always be straightforward, and thus obtaining a thorough patient history is essential. Because triceps tendon ruptures are rare in the general population, a history of strenuous elbow extension activities, steroid use, and/or metabolic disorder should provide clues to the diagnosis. The most common mechanism of injury is a sudden eccentric triceps contraction that can occur during participation in a sport, weight lifting, and falls onto an outstretched arm.

Pain and swelling over the distal triceps are the most common presenting symptoms. As the pain subsides, patients often report elbow weakness and loss of function. Younger patients may report that they are unable to perform pressing exercises at the gym, whereas older patients may lose the ability to push up from a seated position.

Physical Examination

In the acute injury phase, physical examination reveals swelling, ecchymosis, and tenderness near the triceps insertion. In more chronic injuries, a tendon defect may be palpable along with elbow extension weakness compared with the contralateral arm. Active elbow extension against gravity can help determine the degree of tendon tear, but it is important to recognize that patients with complete tears may have preserved elbow extension as the result of an intact lateral tendon expansion. Furthermore, a thorough examination of the skin, arm, and forearm compartments and determination of distal neurovascular status is important to rule out tendon lacerations, compartment syndrome, and ulnar neuropathy, all of which have been associated with distal triceps brachii tendon ruptures.^{59,60}

Imaging

A standard elbow radiographic trauma series (i.e., anteroposterior, lateral, and oblique views) should be ordered to evaluate for bone spurs, bone avulsions, and associated injuries. MRI is an excellent modality for assessment of the integrity of the distal

triceps tendon, especially when determining the extent of the tear. MRI may help distinguish between partial and complete tears and determines the extent of retraction in complete tear injuries. Ultrasound has recently demonstrated the ability to accurately identify partial tears of the distal triceps and reliably differentiate them from complete injuries.⁶¹

Decision-Making Principles

The decision to treat distal triceps tendon ruptures operatively or nonoperatively is based on three factors: (1) the characteristic of the rupture, (2) the patient's medical status, and (3) the patient's functional expectations.

Tear characteristics can be classified using the information in Box 62.3. Partial tears that involve less than 50% of the tendon can often be managed conservatively in patients with adequate active elbow extension. Conservative management of partial tears that involve more than 50% of the tendon and complete tears will typically yield unsatisfactory results, especially in younger, more active patients. In such cases, early primary repair is indicated.⁵⁹ Tear location is also an important consideration. Insertional tears allow for more secure surgical fixation than musculotendinous or muscle belly tears. In proximal tear patterns, a more conservative treatment strategy may be appropriate. Tear chronicity should be determined in all patients. In patients for whom surgery is indicated, all attempts should be made to repair the triceps within 3 weeks⁵⁹; however, primary repair has been reported with success up to 8 months after the time of injury.⁶² An intact lateral expansion may allow for delayed primary repair by preventing proximal retraction and extensive formation of adhesions.

The patient's general medical condition and functional expectations will also guide treatment decisions. Two distinct patient populations are at risk for distal triceps ruptures: first, young, highly functioning athletes, and second, debilitated patients with metabolic disease. A lower threshold for surgery should be afforded to young athletes. Elderly debilitated patients often respond well to conservative treatment as long as they maintain some active extension and have enough power to rise from a seated position.

Treatment Options

Nonoperative Treatment

Nonoperative treatment of distal triceps ruptures consists of 4 weeks of splint immobilization at 30 degrees of elbow flexion. At 4 weeks, ROM exercises are started, and strengthening exercises are added after 8 weeks. The outcomes after conservative treatment are not well documented. There is a paucity of evidence to guide nonoperative care, and the results vary widely. Tendon gapping is an important factor to consider when treating patients nonoperatively. Biomechanical studies reveal a 40% loss of extension strength with as little as 2 cm of gapping.⁶³ Thus in patients in whom significant retraction is observed, nonoperative treatment will likely lead to greater loss of strength and endurance^{64–66} and should be considered only in elderly, sedentary patients and persons too ill to undergo surgery. Muscle belly tears are a distinct entity and heal with scar tissue. Although surgical fixation has been reported in high-level athletes, outcomes are not believed to be altered by surgical intervention.⁶⁷

Operative Treatment

The primary goals of surgical treatment are to reduce the gap between the distal triceps and the olecranon, allow healing to occur, and restore normal strength and endurance to the elbow. Primary repair is optimal and can be accomplished in patients who are treated within 3 weeks of the injury.⁵⁹ Primary repair may be possible after this time window, but reconstructive options should be available as a backup.

Primary repair should be performed by passing a heavy, non-absorbable suture through the triceps tendon with use of either a Bunnell or Krackow whipstitch. Fixation most commonly is achieved by passing the sutures through drill holes in the olecranon and tying them over a bone bridge (Fig. 62.8). Alternatively, knotless suture anchors may be used to fix the tendon to bone. A recent biomechanical study found less displacement after cyclic

loading at the footprint when knotless suture anchors were used in lieu of a traditional transosseous repair with manually tied knots.⁶⁸

Chronic tendon ruptures often require reconstruction, and several options have been described in the literature. An anconeus rotation flap is an excellent option when adequate tissue bulk and quality are available. In one study, however, it was found that this technique was only possible in one of nine cases owing to local tissue quality.⁵⁹ The second option is triceps augmentation with either an autograft or allograft. The most commonly used grafts are the palmaris longus and semitendinosus.^{59,69} Achilles tendon allografts have also been used. Tissue augmentation is achieved by weaving the graft through the remaining triceps tendon, often in a Bunnell fashion, and then attaching it to the olecranon through drill holes (Fig. 62.9).

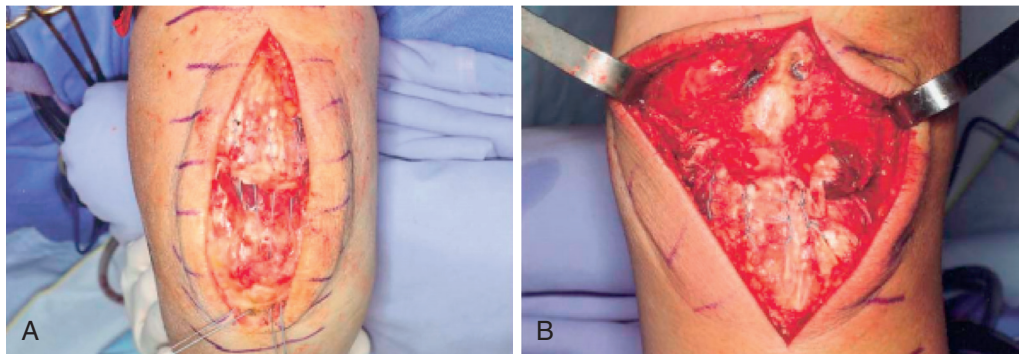


Fig. 62.8 A primary repair using a Krackow suturing technique attached to the olecranon via three drill holes. Sutures are then tied over a bone bridge. (A) Photograph demonstrating suture placement prior to tying. (B) Photographs after suture limbs have been tied over a bone bridge. (From Sierra RJ, Weiss NG, Shrader MW, et al. Acute triceps ruptures: case report and retrospective chart review. *J Shoulder Elbow Surg.* 2006;15[1]:130–134.)

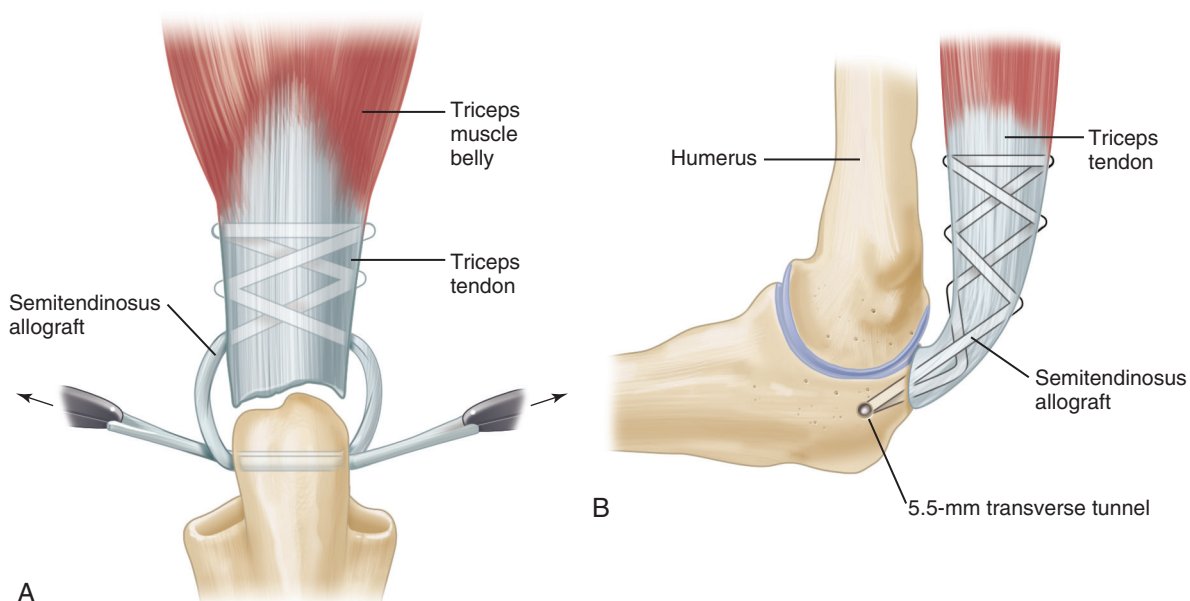


Fig. 62.9 A reconstruction technique using a tendon graft woven through remaining triceps tendon (A) and attached through olecranon drill holes (B). (From Yeh PC, Dodds SD, Smart LR, et al. Distal triceps rupture. *J Am Acad Orthop Surg.* 2010;18[1]:31–40.)

Authors' Preferred Technique

Distal Triceps Tendon Rupture

Recent studies have delineated the anatomy of the distal triceps tendon^{58,70} and described an insertional footprint on the olecranon similar to the footprint of the rotator cuff. The footprint measures 466 mm.³ Information gained from these anatomic studies has led to a new repair technique called the "suture bridge." To perform this technique, a locking Krackow suture is placed through the ruptured triceps tendon, entering and leaving the tendon roughly 2 cm from its distal tip. The olecranon footprint is then identified and excoriated. Two suture anchors are placed at the proximal medial and lateral footprint. Mattress sutures from both anchors are then placed 2 cm from the tendon edge and tied. The sutures from both horizontal mattresses and from the Krackow suture are configured to form a suture bridge and locked into place using two more press-fit suture anchors positioned at the distal end of the footprint (Fig. 62.10).⁷⁰ Care is taken to ensure that none of the anchors penetrates the ulnotrochlear joint. The sutures are secured with the arm held in 35 to 40 degrees of flexion.

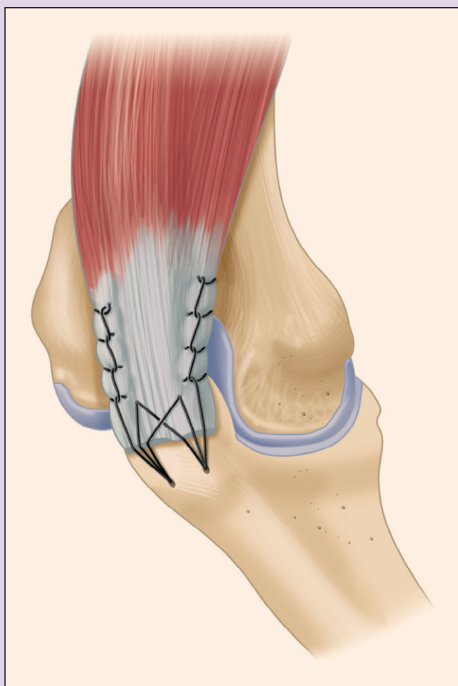


Fig. 62.10 The suture bridge technique. (From Yeh PC, Dodds SD, Smart LR, et al Distal triceps rupture. *J Am Acad Orthop Surg*. 2010;18[1]:31–40.)

Postoperative Management

A standardized postoperative protocol has not been established in the literature. The elbow is often immobilized in a splint at 30 to 40 degrees of flexion for 2 weeks after surgery. At 2 weeks, active-assisted ROM is started if the repair is deemed stable. Some surgeons use a hinged brace to block terminal elbow flexion initially, which is meant to protect the healing tendon from high tensile forces. Light strengthening exercises with a 5-lb weight restriction are started at 6 weeks and progressive resistance exercises, which are best performed under the guidance

of a physical therapist, may begin at 8 to 10 weeks. Return to competitive sport or heavy weight lifting is usually prohibited for 4 to 6 months after the repair is performed.

Results

Van Riet et al.⁵⁹ evaluated 23 patients, including 13 primary repairs via bone tunnels and 9 reconstructions using a variety of techniques. ROM averaged 8 to 138 degrees of flexion after primary repairs and 13 to 133 degrees after reconstructive procedures. All patients regained 4/5 or 5/5 strength on manual muscle testing. Peak strength averaged 92% of the contralateral extremity in the persons who underwent primary repairs and 66% in the persons who underwent reconstructive procedures. These investigators concluded that early surgical repair (i.e., within 3 weeks) is the optimal treatment for distal triceps tendon ruptures.⁵⁹ Yeh et al.⁷¹ performed a biomechanical study evaluating three repair techniques: cruciate repair, suture anchor repair, and anatomic repair (our preferred technique). Load at yield and peak load were similar in all groups; however, the anatomic repair technique most accurately reproduced the anatomic footprint and demonstrated the least amount of displacement after cyclic loading.⁷¹ Recent data have demonstrated excellent results in high demand individuals. Balazs et al.⁷² demonstrated a 94% return to active duty in the military population. In active players in the National Football League, Feinstein et al.⁵⁷ demonstrated a 100% return to play at an average of 165 days.

Complications

In the study of 23 distal triceps brachii repairs by van Riet et al.,⁵⁹ complications included one case of ulnar neuropathy, which resolved; one case of prominent hardware that required removal; and repeat tendon rupture in three patients. Balazs et al.⁷² demonstrated a retear rate of 12.5% in the military personnel, all of which occurred within the first four months following surgery. Other potential complications include olecranon bursitis and elbow flexion contracture.

Future Considerations

The treatment of distal biceps and triceps tendon ruptures has evolved considerably. As our understanding of insertional anatomy and biomechanics has improved, our surgical techniques have improved as well, allowing for stronger fixation methods, improved outcomes, and fewer complications. Early anatomic repair is indicated in healthy, active patients who do not wish to restrict their activities or lose arm strength. Conservative treatment may be appropriate in select cases and may yield satisfactory outcomes when maximal elbow and forearm strength is not necessary or desired. Because the length of recovery is relatively prolonged after these major tendon injuries, future research will likely focus on preventative strategies and methods to decrease tendon-to-bone healing time when injury does occur. The arena of biologic augmentation for optimal tendon healing continues to grow, and translational research will help elucidate the effects of these substances after injury or surgical treatment.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Morrey BF, Askew LJ, An KN, et al. Rupture of the distal tendon of the biceps brachii: a biomechanical study. *J Bone Joint Surg Am.* 1985;67:418–421.

Level of Evidence:

III

Summary:

This classic article presents the biomechanical results of immediate reattachment versus conservative treatment for distal biceps brachii rupture. At 1-year follow-up, it was found that immediate reattachment restored normal strength to the elbow in all patients. Patients treated conservatively lost a mean of 40% of forearm supination strength and 30% of elbow flexion strength.

Citation:

Baker BE, Bierwagen D. Rupture of the distal tendon of the biceps brachii: operative versus non-operative treatment. *J Bone Joint Surg Am.* 1985;67:414–417.

Level of Evidence:

III

Summary:

Baker and Bierwagen performed a case-control study comparing patients treated with surgery versus patients treated conservatively after a distal biceps tendon rupture. Using a Cybex isokinetic dynamometer, it was determined that the group treated conservatively had a 40% reduction in forearm supination strength, a 79% reduction in supination endurance, and a 30% reduction in elbow flexion strength and endurance compared with the group that underwent surgery.

Citation:

van Riet RP, Morrey BF, Ho E, et al. Surgical treatment of distal triceps ruptures. *J Bone Joint Surg Am.* 2003;85:1961–1967.

Level of Evidence:

IV

Summary:

Primary repair of distal triceps tendon ruptures can always be achieved when surgery is performed within 3 weeks. After repair, isokinetic testing showed peak strength of the involved extremity to be 82% and average endurance to be 99% that of the uninvolved extremity. Results of early repair are favorable compared with results of late reconstruction.

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